MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

# Surgical Antimicrobial Prophylaxis Recommendations

## I. INTRODUCTION

The use of peri-operative antimicrobials has become an essential component of the standard of care for certain surgical procedures and can result in a reduced risk of post-operative infection when sound and appropriate principles are utilized. However, the benefit of antimicrobial prophylaxis must be weighed against the risks of toxic and allergic reactions, emergence of resistant bacteria, drug interactions, super-infection, and cost.

## II. CONSIDERATIONS FOR ANTIMICROBIAL PROPHYLAXIS

**Goal:** Administer antimicrobial prophylaxis to achieve serum and tissue levels of antimicrobial at the time of incision and for the duration of the operation, that are in excess of the minimum inhibitory concentration (MIC) needed for organisms that may be encountered during the procedure.

a. Antimicrobial prophylaxis should be administered if there is a risk of infection in the absence of a prophylactic agent; clean procedures rarely require prophylaxis unless high risk procedure, including implantation of prosthetic material.
    i. Clean procedures are defined as those with no acute inflammation or transection of gastrointestinal, oropharyngeal, genitourinary, biliary, or respiratory tracts (elective cases, no technique break).
b. The activity of the chosen prophylactic agent(s) should encompass the most common pathogens associated with the surgical procedure and consider local susceptibility data, but need not cover every likely pathogen.
c. The prophylactic agent must be administered in a dose which provides an effective tissue concentration prior to incision / intra-operative bacterial contamination.
    i. In most instances, a single intravenous dose of an antimicrobial agent provides adequate tissue concentrations around the time of anesthesia induction and throughout the operation.
        1. Antimicrobial agent infusion should begin 15-60 minutes before the incision with the exception of vancomycin, levofloxacin, ciprofloxacin, gentamicin, azithromycin and fluconazole. These infusions should begin 45-90 minutes before the incision and infused over 60-120 minutes as indicated for adults and pediatrics (See following tables**).**
        2. In adult patients, cefazolin (2 g if < 120 kg, 3 g if ≥ 120 kg) and vancomycin (1 g if < 80 kg, 1.5 g if ≥ 80 kg) dosing is based on weight. Adult patients < 50 kg should refer to Patients < 50 kg (Adult and Pediatric) Dosing recommendations for dosing. Weight-based dosing is recommended for all antibiotics in patients < 50 kg.
        3. Infusion duration and time to redosing for recommended prophylactic antimicrobials are summarized for adults and pediatrics.
        4. **All prophylactic antimicrobials should be discontinued after the intra-operative period, unless otherwise specified.**
            a. Data have not supported subsequent doses after surgical closure and may increase the risk of *Clostridium difficile* and antimicrobial resistance.

A longer duration of antimicrobials may be indicated, if concomitant infection is present at the time of surgery

| CARDIOTHORACIC PROCEDURES | | | |
|---|---|---|---|
| Nature of operation | Likely pathogens | Recommended regimen[α] – | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
| **Cardiac surgery with implants**<br><br>**Congenital heart repair procedures requiring an open sternum postoperatively**<br><br>**Pacemaker or AICD placement or revision**<br><br>**Heart transplant**<br><br>**Left Ventricular Assist Device (LVAD)** | *S. aureus*<br><br>*S. epidermidis*<br><br>gram-negative bacilli | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin** |

## GASTROINTESTINAL PROCEDURES

| Nature of operation | Likely pathogens | Recommended regimen[α] | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
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| **Appendectomy**<br>(non-perforated) | Enteric gram-negative bacilli<br><br>Enterococci<br><br>anaerobes (Bacteroides spp., Clostridia)<br><br>*S. aureus* | Pediatrics:<br>**Cefoxitin**<br>OR<br>**Piperacillin-tazobactam** | Pediatrics:<br>**Clindamycin**<br>+ **Gentamicin** |
| **Colon and anorectal** procedures NOT requiring antibiotic prophylaxis:<br>• Evaluation under anesthesia, fulguration of warts, high resolution anoscopy, dilation of stricture, anal biopsy<br>• EUA for fistula placement of seton<br>• Fistulotomy, simple (NOT a Surgisis plug or advancement flap),<br>• Transanal resection of fibroepithelial or pedunculated polyp | None | Prophylaxis Not Recommended | Prophylaxis Not Recommended |
| **Splenectomy** | *S. aureus*<br>*S. epidermidis* | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Cefuroxime** OR<br>**Vancomycin** |
| **Esophageal, gastric surgery, G-tube; peg tube**<br>(See small bowel for gastric bypass and gastrectomy) | Upper airway flora (aerobic, anaerobic Streptococci)<br><br>*S. aureus*<br>more rarely aerobic gram-negative bacilli | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin**<br>OR<br>**Vancomycin** |

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|---|---|---|---|
| **Colorectal Procedures**<br>Not limited to, but including the following:<br>• Colon & rectal resection<br>• High, complex fistula by Surgisis plug or advancement flap<br>• Lateral sphincterotomy for anal fissure<br>• Hemorrhoidectomy<br>• Transanal resection for sessile polyp, villous adenoma, possible T1 malignancy<br>• Rectal prolapse procedure (Altmeier or abdominal rectosigmoid resection/rectopexy)<br>*(Some procedures may be included in SCIP, and appropriate antibiotic selection is linked to hospital reimbursement)* | Enteric gram-negative bacilli<br><br>Enterococci<br><br>anaerobes (Bacteroides spp., Clostridia)<br>*S. aureus* | Pediatrics:<br>**Cefoxitin**<br><br>Optional oral regimens:<br>**Neomycin** 20 mg/kg/dose PO<br>+ **Metronidazole** 10 mg/kg/dose PO OR<br>**Neomycin** 20 mg/kg/dose<br><br>+<br>**Erythromycin**<br>base 20<br>mg/kg/dose | Pediatrics:<br>**Cefazolin**<br>+ **Metronidazole**<br>OR<br>**Clindamycin**<br>+ **Gentamicin** |
| **Hepatic, biliary tract, pancreatic including cholecystectomy and gallbladder procedures, cystgastrostomy**<br>**(see separate section for Pancreaticoduodenectomy)**<br><br>(excluding low-risk laparoscopic cholecystectomy*) | Enteric gram-negative bacilli (e.g., *E. coli*, *Klebsiella*)<br><br>Enterococci<br><br>*S. aureus*<br>anaerobes (*Bacteroides spp.*, *Clostridia*)<br>common with stents, biliary obstruction | Pediatrics:<br>**Cefoxitin**<br>OR<br>**Piperacillin-tazobactam** (for biliary atresia repair) | Pediatrics:<br>**Clindamycin**<br>+ **Gentamicin** |
| **Hernia repair**<br>(hernioplasty-prosthetic mesh repair of hernia; herniorrhaphy-suture repair of hernia) | *S. aureus*<br>*S. epidermidis* | Pediatrics:<br>**Cefazolin** | Pediatrics: **Clindamycin** OR<br>**Vancomycin** |
| **Small bowel, gastric bypass, gastrectomy** | Enteric gram-negative bacilli Enterococci<br><br>anaerobes (Bacteroides spp., Clostridia)<br>*S. aureus* | Pediatrics:<br>**Cefoxitin**<br>OR<br>**Piperacillin-tazobactam** (for biliary atresia repair) | Pediatrics:<br>**Clindamycin**<br>+ **Gentamicin** |

## GENITOURINARY PROCEDURES

**Prophylaxis of genitourinary procedures that enter the urinary tract should include coverage of likely pathogens as well as organisms from the most recent urologic source (urine culture, stone culture, etc) within the preceding year, regardless of prior antimicrobial treatment. If the most recent positive urologic source culture was > 1 year ago, standard prophylaxis should be utilized**

| Nature of operation | Likely pathogens | Recommended regimen[α] | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
|---|---|---|---|
| **Consider Prophylaxis ONLY in patients with risk factors:**<br>Cystography, urodynamic study, simple cystourethroscopy, shock wave lithotripsy<br>Risk Factors Include<br>• Poor functional status/frailty<br>• anatomic anomalies of urinary tract<br>• chronic steroid use<br>• immunocompromising condition or recent systemic chemotherapy | Enteric gram-negative bacilli | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Cefoxitin**<br>OR<br>**Gentamicin**<br>± **Ampicillin**<br>OR<br>**TMP-SMX** (Bactrim®) |
| **Prophylaxis recommended for:**<br>Ureteroscopy ± stent placement<br>Cystourethroscopy with manipulation including:<br>• transurethral resection of bladder tumor and prostate,<br>• any biopsy, resection, fulguration, foreign body removal, urethral dilation or urethrotomy<br>• any ureteral instrumentation including catheterization or stent placement/removal<br>• submucousal injection (e.g., Botulinum toxin)<br><br>*The following procedures do not require infusion of IV antibiotics timed for 15-30 min prior, but can be given directly before the procedure, as they are intended for prevention for post-procedural UTI: bladder biopsy; botox injection; cystolithalopaxy; cystoscopy with fulguration, microplastique, bladder neck incision or retrograde pyelogram; hydrodistention; ureteral stent placement and removal; ureteroscopy. | *E. coli*<br>*Proteus* spp.<br>*Klebsiella* spp. | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Cefoxitin**<br>OR<br>**Gentamicin**<br>± **Ampicillin**<br>OR<br>**TMP-SMX** (Bactrim®) |

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|---|---|---|---|
| **Circumcision** | Staphylococci | Pediatrics:<br>No antibiotic prophylaxis in healthy neonates; otherwise:<br>**Cefazolin** or Amoxicillin | Pediatrics:<br>**Clindamycin** |
| **Genitourinary procedures involving small or large intestine**<br><br>Including urinary diversions, cystectomy with small bowel conduit, uretero-pelvic junction repair, colon conduits, radical cystectomy neobladder/ileostomy, etc | *S. aureus*<br>*S. epidermidis*<br>Streptococci<br>Enteric gram-negative bacilli | Pediatrics:<br>**Cefoxitin** | Pediatrics:<br>**Ampicillin-sulbactam**<br>OR<br>**Clindamycin**<br>+ **Gentamicin** |
| **Percutaneous nephrolithotomy (PCNL)**<br><br>High risk features include:<br>• Positive urine culture within 2-4 weeks<br>• Residual stone with prior positive stone culture<br>• Current indwelling ureteral stent or nephrostomy tube<br>• Severe hydronephrosis<br>• Continuous intermittent catheterization<br>• Renal transplant or other severe immunocompromising condition<br>• Neurogenic bladder (with or without urinary diversion)<br>• Urinary Diversion<br><br>Chronic indwelling catheters (Foley or SP tube) | *S. aureus*<br>*S. epidermidis*<br>enteric gram-negative bacilli | Pediatrics:<br>**Cefazolin** | Pediatrics :<br>**Clindamycin**<br>+ **Gentamicin** |
| **Implanted prosthetic devices**<br><br>Artificial urinary sphincter and sacral nerve stimulators | Staphylococci<br>enteric gram-negative bacilli | Pediatrics:<br>**Cefazolin**<br>+ **Gentamicin** | Pediatrics:<br>**Cefoxitin**<br>OR<br>**Ampicillin-sulbactam**<br>OR<br>**Vancomycin**<br>+ **Gentamicin** |

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| Nature of operation | Likely pathogens | Recommended regimen[α] | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
|---|---|---|---|
| **Vaginal urologic surgery**<br>Including urethral sling, fistulae repair, urethral diverticulectomy, urethropexy<br><br>**Open or laparoscopic procedure involving entry into the urinary tract**<br>Including urethroplasty; stricture repair including urethrectomy. Nephrectomy, partial or otherwise, urethrectomy, pyeloplasty, radical prostatectomy, partial cystectomy | Enteric gram-negative bacilli.<br><br>*S. aureus*<br><br>*S. epidermidis*<br><br>*Streptococci spp.* (vaginal surgery mostly) | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Cefoxitin**<br>OR<br>**Ciprofloxacin** |
| **Inguinal and scrotal cases**<br><br>Including radical orchiectomy, reversals, variocelectomy, hydrocelectomy | *Staphylococci spp.*<br><br>Enteric gram-negative bacilli | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin** |

| HEAD AND NECK PROCEDURES | | | |
|---|---|---|---|
| Nature of operation | Likely pathogens | Recommended regimen[α] | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
| See: [link] | | | |

| NEUROSURGICAL PROCEDURES | | | |
|---|---|---|---|
| Nature of operation | Likely pathogens | Recommended regimen[α] | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
| **Craniotomy**<br>**VP shunts & other prosthetic material**<br>**Spine implantable devices**<br>*(Some procedures may be included in SCIP, and appropriate antibiotic selection is linked to hospital reimbursement)* | *S. aureus*<br>*S. epidermidis* | Pediatrics:<br>**TMP-SMX** (Bactrim ™) | Pediatrics:<br>**Cefazolin**<br>OR<br>**Vancomycin** |
| **Discography**<br>(Intradiscal antimicrobial prophylaxis is not endorsed by the UMHS Pharmacy and Therapeutics Committee and should not be used.) | *S. aureus*<br>*S. epidermidis* | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin**<br>OR<br>**Vancomycin** |
| **Skull base, anterior approach including transphenoidal surgery for pituitary tumors** | *S. aureus*<br>*S. epidermidis*<br>gram-negative bacilli | Pediatrics:<br>**Clindamycin**<br>+ **Cefuroxime** | Pediatrics:<br>**Vancomycin** |
| **Skull base, lateral or posterior approach procedures** | *S. aureus*<br>*S. epidermidis* | Pediatrics:<br>**Clindamycin**<br>+ **Cefuroxime** | Pediatrics:<br>**Vancomycin** |

| ORTHOPEDIC PROCEDURES | | | |
|---|---|---|---|
| Nature of operation | Likely pathogens | Recommended regimen[α] | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
| **Total joint replacement** (Arthroplasty)<br><br>**Implantation of prosthetic material** (e.g., intramedullary nails, screw, plates, wires)<br><br>**Hip fracture repair** | *S. aureus*<br><br>*S. epidermidis* | Pediatrics:<br>**Cefazolin**<br>History of MRSA carriage or infection:<br>**Vancomycin**<br>+ **Cefazolin** | Pediatrics:<br>*History of MRSA carriage or infection:*<br>**Vancomycin**<br><br>*Without history of MRSA carriage or infection:*<br>**Clindamycin** |
| **Open Fracture Repair**<br>(Includes upper and lower extremity open fractures)<br><br>**NOTE: These are abbreviated recommendations. Please see full guideline on antimicrobial stewardship website (under 'Bone and Joint Infections' header):**<br>**https://antimicrobialstewardship.med.umich.edu/guidelines/adult** | *S. aureus Streptococcus gram-negative rods* | Pediatrics:<br>*Type I or II open fracture:*<br>**Cefazolin**<br>Extensive soil exposure (e.g., farming accident)<br>+ **Metronidazole**<br>*Type III open fracture:*<br>**Ceftriaxone**<br>Extensive soil exposure (e.g., farming accident)<br>+ **Metronidazole**<br>Antibiotic prophylaxis should be provided for 24 hours but may be extended to 48 hours if needed. | Pediatrics:<br>*Type I or II open fracture:*<br>*History of MRSA carriage or infection*<br>**Vancomycin**<br>*Without history of MRSA carriage or infection:*<br>**Clindamycin**<br>*Type III open fracture:*<br>*History of MRSA carriage or infection*<br>**Vancomycin**<br>+ **Aztreonam**<br>*Without history of MRSA carriage or infection:*<br>**Clindamycin**<br>+ **Aztreonam**<br>Antibiotic prophylaxis should be provided for 24 hours but may be extended to 48 hours if needed. |
| **Spinal procedure**, with or without instrumentation | *S. aureus*<br><br>*Coagulase negative staphylococci*<br>*Gram negative bacilli* | Pediatrics:<br>**Cefazolin**<br><br>History of MRSA carriage or infection:<br>**Vancomycin**<br>+ **Cefazolin** | Pediatrics:<br>*History of MRSA carriage or infection*<br>**Vancomycin**<br><br>*Without history of MRSA carriage or infection:*<br><br>**Clindamycin** |

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| **Sports Medicine**<br>(Orthopaedic Surgeries)<br>**Implantation of interference screws, suture anchors, permanent sutures, etc.** | *S. aureus Streptococcus* | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin**<br>OR<br>**Vancomycin** |

| **PLASTIC SURGERY PROCEDURES** | | | |
|---|---|---|---|
| Nature of operation | Likely pathogens | Recommended regimen[α] | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
| **Breast reconstruction** (without implants), **cosmetic procedures** (excluding blepharoplasty), **large hand dissections*, phalloplasty, vaginoplasty, and general reconstructive procedures involving medium/large flaps or tissue expanders**<br><br>*Open fractures: see Orthopedic Surgery guidelines | *S. aureus*<br><br>*S. epidermidis* | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin** |
| **Breast procedures with implants** | *S. aureus*<br><br>*S. epidermidis* | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin** |
| **Cleft lip** and **palate repair,** or **facial procedures** that transect **oral, nasal (see rhinoplasty below), or pharyngeal mucosa** | Oral anaerobes<br><br>enteric gram-negative bacilli<br><br>*S. aureus*<br><br>*S. epidermidis*<br><br>viridans streptococci | Pediatrics:<br>**Ampicillin-sulbactam** | Pediatrics:<br>**Clindamycin** |
| **Rhinoplasty/Septorhinoplasty** | *S. aureus*<br><br>*S. epidermidis*<br><br>*C. acnes* | Pediatrics:<br>**Ampicillin-sulbactam**<br>OR<br>**Cefazolin** | Pediatrics:<br>**Clindamycin** |

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| **Open cranial vault reconstruction** | *S. aureus*<br>*S. epidermidis*<br>Oral flora<br>*P. aeruginosa* | Pediatrics:<br>Pre-operative:<br>Piperacillin-tazobactam<br>Post-operative:<br>Ampicillin-sulbactam | Pediatrics:<br>Pre-operative:<br>Clindamycin<br>+ Gentamicin<br>OR<br>Clindamycin<br>+ Aztreonam<br><br>Post-operative:<br>Clindamycin |
| **Endoscopic craniofacial procedures and strip craniectomy** | *S. aureus*<br>*S. epidermidis* | Pediatrics:<br><br>Cefazolin | Pediatrics:<br><br>Clindamycin |

| RADIOLOGY PROCEDURES | | | |
|---|---|---|---|
| Nature of operation | Likely pathogens | Recommended regimen[α] | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
| **Any ARTERIAL OR VENOUS intervention that involves the hepatobiliary or reproductive systems.**<br>• Hepatic arterial embo for hepatic trauma, suspected hemobilia, Y-90 treatment (not mapping), TACE, or bland embo.<br>• Any hepatic arterial embo in a transplanted liver or any liver at high-risk for ischemia (occluded/stenotic portal vein, etc.).<br>• TIPS/DIPS Placement<br>• BRTO/BATO/PARTO/CARTO<br>• Portal vein embolization<br>• Uterine artery embo<br>• IVC filter retrieval with bowel penetration<br><br>**Biliary Interventions**<br>• PTC<br>• Biliary drain check/change<br>• Biliary stent placement<br>• Cholecystostomy tube placement<br>• Cholecystostomy tube check/change<br>• Gallbladder cryo- and chemical ablation<br>• Biliary endoscopy<br><br>**Thermal ablation of the liver** | Skin Flora: *S. aureus, S. epidermis*<br><br>GI Flora: Enteric gram-negative bacilli, anaerobes | Pediatrics:<br>**Cefoxitin** | Pediatrics:<br>**Clindamycin**<br>+ **Gentamicin**<br><br>*Alternative if any allergy to gentamicin **OR** SCr 2 mg/dL **OR** CrCl < 40 mL/min:*<br>**Clindamycin**<br>+ **Aztreonam** |

| RADIOLOGY PROCEDURES | | | |
|---|---|---|---|
| Nature of operation | Likely pathogens | Recommended regimen[α] | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
| **Any ARTERIAL EMBOLIZATION involving the GU system**<br>• Prostate artery embolization<br>• Renal embolization with > 70% of the kidney expected to be embolized.<br><br>**GU Interventions**<br>• PCN placement or check/change/conversion<br>• NUS placement or check/change/conversion<br>• UDC placement or check/change/conversion<br><br>**Thermal ablation of the kidney with treatment zone including the collecting system**<br><br>**Suprapubic Tube Placement** | Skin Flora: *S. aureus, S. epidermis*<br>GU Flora: *E. coli, Proteus, Klebsiella* | Pediatrics:<br>**Cefoxitin** | Pediatrics:<br>**Clindamycin**<br>+ **Gentamicin**<br><br>*Alternative if any allergy to gentamicin **OR** SCr 2 mg/dL **OR** CrCl < 40 mL/min*<br>**Clindamycin**<br>+ **Aztreonam** |
| **Renal Arterial Embolization with < 70% of the kidney expected to be embolized** | Skin Flora: *S. aureus, S. epidermis*<br><br>GU Flora: *E. coli, Proteus, Klebsiella* | **NOT USUALLY INDICATED;** consider in the following instances:<br>• Immunosuppressed<br>• Plan to leave sheath in place<br><br>Pediatrics:<br><br>**Cefoxitin** | Pediatrics:<br>**Clindamycin**<br>+ **Gentamicin**<br><br>*Alternative if any allergy to gentamicin **OR** SCr 2 mg/dL **OR** CrCl <40 mL/min:*<br>**Clindamycin**<br>+ **Aztreonam** |
| **Any ARTERIAL, VENOUS, OR DIALYSIS angio or intervention with plan for covered stent placement, overnight lysis, or plan to leave sheath in place.**<br>• Any of these interventions that may result in placement of a covered stent, overnight lysis, or plan to leave the sheath in place.<br>• Distal splenic artery embolization for trauma or aneurysm with > 70% of the spleen expected to be embolized<br>• Partial Splenic Embolization for hypersplenism<br>• AVF or AVG Declots<br>**Lung Ablation PleurX Placement** | Skin Flora: *S. aureus, S. epidermis* | Pediatrics:<br>**Cefazolin**<br><br>*If history of MRSA infection or colonization:*<br>**Vancomycin** | Pediatrics:<br>**Vancomycin**<br>OR<br>**Clindamycin** |

| RADIOLOGY PROCEDURES | | | |
|---|---|---|---|
| Nature of operation | Likely pathogens | Recommended regimen[α] | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
| **Any ARTERIAL, VENOUS, OR DIALYSIS angio or intervention with NO plan for covered stent placement, overnight lysis, or plan to leave sheath in place.**<br>• Arterial, venous, or dialysis surveillance, recanalization, angioplasty, and bare metal stent placement.<br>• Any arterial embolization not involving a solid organ. Includes bronchial artery embo, GDA embo, Y-90 mapping (not treatment), GI bleed embo (not if hemobilia is suspected), bariatric embo, genicular artery embo, or other visceral artery embo.<br>• Proximal splenic artery embo<br>• Distal splenic artery embo or splenic artery aneurysm embo with < 70% of the spleen expected to be embolized<br>• IVC filter placement and IVC filter retrievals with NO bowel penetration.<br>• Superficial Vein Ablation / Sclerotherapy<br>• TIPS/DIPS check (no stent extension or relining).<br>• Embo or sclero of vascular anomaly not involving the hepatobiliary, splenic, renal, GI, or GU systems.<br><br>**Line placements and exchanges**<br><br>**Thermal ablations of bone or kidney (treatment zone cannot include the collecting system)**<br><br>**Gastrostomy/gastrojejunostomy placement/exchange, Jejunostomy exchanges**<br><br>**Paracentesis, Thoracentesis**<br>• **Pain interventions** not requiring needle passage through bowel, liver, or kidneys. | Skin Flora: *S. aureus, S. epidermis* | **NOT USUALLY INDICATED;** consider in the following instances:<br>• Immunosuppressed<br>• Plan to place covered stent, overnight lysis, or plan to leave sheath in place<br><br>Pediatrics:<br>**Cefazolin**<br><br>*If history of MRSA infection or colonization:*<br>**Vancomycin** | Pediatrics:<br>**Vancomycin**<br>OR<br>**Clindamycin** |

| SOLID ORGAN TRANSPLANT PROCEDURES | | | |
|---|---|---|---|
| Nature of operation | Likely pathogens | Recommended regimen[α] | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
| **Kidney transplant** | *S. aureus*<br>*S. epidermidis*<br>enteric gram-negative bacilli | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Vancomycin**<br>+ **Aztreonam** |
| **Liver transplant** | Enteric gram-negative bacilli (e.g., *E. coli*, *Klebsiella*)<br>Enterococci<br>S. *aureus*<br>anaerobes (*Bacteroides, Clostridia*) | Pediatrics:<br>**Piperacillin-tazobactam** | Pediatrics:<br>**Vancomycin**<br>+ **Aztreonam** |
| For **Heart Transplant** see Cardiothoracic guideline<br>For **Lung Transplant** see Thoracic (non-cardiac) guideline | | | |

| THORACIC (NON-CARDIAC) PROCEDURES | | | |
|---|---|---|---|
| Nature of operation | Likely pathogens | Recommended regimen[α] | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
| **Esophagectomy** | *S. aureus*<br>*S. epidermidis* | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin**<br>OR<br>**Vancomycin** |
| **Misc. thoracic procedures** | *S. aureus*<br>*S. epidermidis* | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin**<br>OR<br>**Vancomycin** |

| VASCULAR PROCEDURES | | | |
|---|---|---|---|
| Nature of operation | Likely pathogens | Recommended regimen[α] | Alternative regimen[α]<br>See footnote for optimal approach in patients with β-lactam allergies: **almost all patients with penicillin allergies, including anaphylaxis, can receive cephalosporin-based prophylaxis** |
| **Open, aortic aneurysm repair**<br><br>**Aortic and peripheral aneurysm repair with endovascular stent graft**<br><br>**Arterial bypass with prosthetic graft** | *S. aureus*<br>*S. epidermidis* | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin**<br>OR<br>**Vancomycin** |
| **AV grafts** (with prosthetic) & **fistulas** (no prosthetic) **with skin flap for vein transposition** | *S. aureus*<br>*S. epidermidis* | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin** OR<br>**Vancomycin** |
| **Carotid endarterectomy with prosthesis or patch** | *S. aureus*<br>*S. epidermidis* | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin** OR<br>**Vancomycin** |
| **Lower extremity amputation for ischemia**<br>**Phlebectomy of varicose veins, stripping of varicose veins, ligation of varicose veins** (e.g., Saphenous Vein stripping or ligation) | *S. aureus*<br>*S. epidermidis*<br>enteric gram-negative bacilli<br>*Clostridia* spp. | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin** OR<br>**Vancomycin** |
| **Thromboendarterectomy without bypass Arterial bypass with vein graft** | *S. aureus*<br>*S. epidermidis* | Pediatrics:<br>**Cefazolin** | Pediatrics:<br>**Clindamycin** OR<br>**Vancomycin** |

[2] Low-risk allergies include: pruritus without rash, remote (>10 years) unknown reaction, patient denies allergy but is on record, mild/severe rash with no other symptoms (see exceptions in 'severe non-IgE reactions below), urticaria/hives with no other symptomsSee β-lactam allergy evaluation and empiric guidance for further information.

[3] High-risk allergies include: respiratory symptoms, angioedema, cardiovascular symptomssyncope/passing out, arrhythmia), anaphylaxis. **If a patient has a high-risk allergy to penicillins, they may receive cefazolin, cefoxitin, or cefuroxime.**. See β-lactam allergy evaluation and empiric guidance for further information.

[4] Severe non-IgE reactions that are contraindications to further beta-lactam use (**except aztreonam, which can be used unless the reaction was to ceftazidime or cefiderocol**) unless approved by Allergy: organ damage (kidney, liver), drug-induced immune-mediated anemia/thrombocytopenia/leukopenia, rash with mucosal lesions (Stevens Johnson Syndrome/Toxic Epidermal Necrosis), rash with pustules (acute generalized exanthematous pustulosis), rash with eosinophils and organ injury (DRESS – drug rash eosinophilia and systemic symptoms), rash with joint pain, fever, and myalgia (Serum Sickness). See β-lactam allergy evaluation and empiric guidance for further information.

[5] Beta-lactam antibiotics include the following antibiotic classes: penicillins, cephalosporins, carbapenems, aztreonam

α Refer to Pre-op and Intraoperative Antibiotic Dosing Recommendations Guidelines

- Guideline includes infusion duration and time to intra-operative redosing
- All prophylactic antimicrobials should be discontinued after the intra-operative period, unless otherwise specified
- Adult patients < 50 kg should receive antibiotic dosing based on ≤ 50 kg guideline
- Pediatrics patients > 50 kg should receive antibiotic dosing based on > 50 kg guideline
- Use mg/kg dosing for patients < 5 kg
- SIP operative pediatrics procedures include: cardiac, ventriculoperitoneal shunts, and spinal surgery

* Redosing should occur with the same initial dose, with the exception of cefazolin and clindamycin when used in cardiac/SBE prophylaxis

** SBE prophylaxis, ACC/AHA 2008 guideline update on valvular heart disease

π Adapted from Clinical Infectious Diseases 2004;38:1706-15 and Am J Health-Syst Pharm 2013;70.

γ Reconstituted dose injected directly into vein or via running IV fluids (only if IV piggyback not available).

δ Intermittent IV infusion.

ε Gentamicin dose should be based on ideal body weight unless actual body weight is less than ideal body weight. Use alternative to gentamicin in adult patients if SCr ≥ 2 mg/dL or CrCL < 40 mL/min, unless otherwise specified.

ζ Daptomycin dosing is based on actual body weight

λ Infusion Timing:

- Infusions should begin 15-60 minutes prior to incision for all antimicrobial agents unless listed below:
  - Levofloxacin, ciprofloxacin, vancomycin, gentamicin, azithromycin, and fluconazole, which should begin 45-90 minutes prior to incision.
- If pre-operative antibiotics have already been administered but incision has been delayed more than 60 minutes beyond the maximum dosing window (i.e., beta-lactam antibiotics started >2 hours prior to incision), consider pre-operative re-dosing for all antibiotics except vancomycin and aminoglycosides. For vancomycin and aminoglycosides, please contact pharmacy to determine if re-dosing is appropriate.

ξ Patients receiving systemic antibiotics prior to procedure should still receive the standard pre-operative antimicrobial prophylaxis with appropriate timing of administration as outlined in the UMHS surgical antimicrobial prophylaxis guidelines. Given the risk of nephrotoxicity, in patients receiving vancomycin or aminoglycosides prior to procedure who need these agents for pre-operative antimicrobial prophylaxis, please consult pharmacy to see if treatment doses can be rescheduled such that administration begins 45-90 minutes prior to incision.

Σ In patients with known colonization or infection with drug-resistant pathogens, the standard pre-operative antimicrobial prophylaxis should generally still be administered unless otherwise specified in procedure-specific guidelines. Please contact the ID approval pager (adults: 30780; pediatrics: 36149) to discuss the case if concerned about: history of a multidrug-resistant organism (e.g., ESBL-producing organism, carbapenem-resistant Enterobacteriaceae, etc.) at the site of the procedure in the previous year OR history of MRSA colonization in the previous year.

Ω Deviations may be appropriate for staged procedures. For example, in TMJ replacement cases ampicillin/sulbactam prior to the intra-oral portion of the procedure followed by cefazolin prior to the neck incision.

| **Last revised:** 03/2025 | **Antimicrobial Stewardship Subcommittee:** 03/2025 | **P&T:** 03/2025 |
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## References


| | Citation |
|---|---|
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| 3 | Bratzler DW., Houch PM. Antimicrobial Prophylaxis for Surgery: An Advisory Statement from the National Surgical Infection Prevention Project. Am J Surg. 2005 Apr;189(4):395-404. |
| 4 | American Academy of Pediatrics: Antimicrobial Prophylaxis in Pediatric Surgical Patients. Pediatrics. 1984 Sep;74(3):437-9. |
| 5 | Bratzler DW, et al. Clinical practice guidelines for antimicrobial prophylaxis in surgery. Am J Health Syst Pharm. 2013 Feb 1;70(3):195-283. |


## Revision History

| Date | Revision |
|---|---|
| 08/2020 | Revised OB/GYN section |
| 10/2020 | Revised Urology section |
| 11/2020 | Revised Gastrointestinal section |
| 03/2021 | Revised allergy wording, Orthopedic, Plastics, and Cardiothoracic sections |
| 04/2021 | Adjusted rhinoplasty/septorhinoplasty and vasectomy recommendations |
| 12/2021 | Revised Orthopedic section |
| 03/2022 | Revised dosing recommendations for patients ≤ 50 kg |
| 10/2022 | Added Radiology, updated allergy wording |
| 11/2022 | Revised pediatric vancomycin, gentamicin/tobramycin dosing |
| 01/2023 | Adjusted allergy wording, revised beta-lactam footnote |
| 03/2023 | Removed redosing recommendation for piperacillin-tazobactam during liver transplant |
| 08/2023 | Revised Gastrointestinal section |
| 02/2024 | Revised Cardiothoracic section |
| 12/2024 | Revised Orthopedic section for open fractures |
| 03/2025 | Replaced vancomycin with daptomycin throughout document; revised daptomycin dosing |

*The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source*